

Financial hardship in Amsterdam

2026-06-23

Set-up your environment

```
f{r package_install, include=FALSE} # install.packages("tidyverse") # install.packages("yaml")  
# install.packages("rmarkdown") # install.packages("ggplot2") # install.packages("dplyr")  
# install.packages(c("sf", "tmap" , "cbsodataR")) # install.packages("tinytex") #tinytex::install_tinyt
```

Title Page

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Tutorial 1 Group 4

Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

Social Problem: People in Amsterdam who struggle to make ends meet.

Many people living in Amsterdam have difficulty covering their daily expenses, such as rent, food and energy bills. This financial pressure can lead to stress, debt, and a lower quality of life.

Why is this relevant?

This issue has become increasingly relevant due to rising housing costs, inflation, and unequal income growth across households. Recent data from the Centraal Bureau voor de Statistiek (CBS, 2025) show that in 2024, 13.3% of Dutch adults reported having difficulty making ends meet, while financial dissatisfaction remains concentrated among lower-income groups. Additionally, 26.1% of adults reported having many financial worries about their future, indicating that financial insecurity remains a structural issue in Dutch society. Research by Mullainathan and Shafir (2013) shows that financial scarcity can lead to chronic stress, reduced decision-making capacity, and lower overall wellbeing. In cities like Amsterdam, where living costs and rent prices have increased substantially over the last two decades, lower-income households are more vulnerable to financial pressure.

Part 2 - Data Sourcing

2.1 Load in the data

2.2 Provide a short summary of the dataset(s)

The dataset `InkomenPerWijk2023Gefilterd` contains all the neighbourhoods of Amsterdam as well as the average income of a household in a year for each of these neighborhoods.

The dataset `moeiteRondkomenAlleJaren` contains codes for all the neighbourhoods of The Netherlands and the amount of families who have difficulty to make ends meet. This data is available for multiple years

The dataset `Wijken_Amsterdam_2024_CBS` links the codes for the neighbourhoods used in dataset `moeiteRondkomenAlleJaren` to the correct neighbourhoods.

The dataset `amsterdam_wijken_nieuw2` contains the codes from dataset `moeiteRondkomenAlleJaren`, as well as the correct neighbourhoods from dataset `Wijken_Amsterdam_2024_CBS` and links it to the amount of families who have difficulty to make ends meet from multiple periods.

The dataset `gemiddeldehuur2023` contains the average rent in a neighbourhood for all neighbourhoods of Amsterdam. This data is available for multiple years. Limitations of the datasets:

This study has several limitations.

First, the datasets are used at neighbourhood level, which means individual household variation cannot be observed. This may lead to a problem where conclusions of individuals are based on group-level data.

Second, the main variable (*Moeite met rondkomen*) is data made from self reports, meaning that subjective interpretation may affect responses.

Third, the student variable may overestimate the actual student population, since not everyone aged 15–25 is enrolled in education.

Finally, important variables such as debt, savings, household composition, education level, and employment type are missing. These factors may also strongly influence financial situations.

2.3 Describe the type of variables included

The variables that are used in the datasets are average rent which is called `huur gemiddeld`, the percentage of people who have difficulty to make ends meet which is called `moeitemetrondkomen_32` and the average rent in a neighbourhood divided by 1000 which is called `Inkomen/Inkomen van huishoudens/Gem. gestandaardiseerd inkomen van huish (x 1 000 euro) (ascending)`. All of these variables are numeric variables and ratios because they all have a true zero point and it is obvious which value is higher.

the variables `Jaar` and `Perioden` both show a date in time `Jaar` is rounded to years while `Perioden` shows years and months. This is a numeric variable and an interval since there is no true zero point but the difference between values is meaningful.

The variables `Wijken en buurten`, `WijkenEnBuurten`, `Wijknaam` and `Naam` all refer to the names of neighbourhoods. These are categorical variables and nominal variables, because they have no natural order and are only categorical variables.

The variables `CBS Wijkcode` and `Codering_3` both refer to codes which are linked to certain neighbourhoods, These are also categorical variables and nominal variables, because they have no natural order and are only categorical variables.

Part 3 - Quantifying

3.1 Data cleaning

Before analysis, the datasets required several cleaning steps to make sure that the graphs could be made without gaps and with consistent data.

First, to start off after importing the raw datasets, all datasets were filtered to include only neighbourhood-level observations for Amsterdam. This improved the focus of the analysis on Amsterdam.

Second, irrelevant variables such as identification numbers, age categories, and margin-of-error variables were removed to simplify the dataset.

Third, string variables such as neighbourhood names and codes were cleaned using `trimws()` to remove unnecessary spaces. This was necessary for merging datasets accurately.

Fourth, duplicate observations were removed using `distinct()`, ensuring that each neighbourhood appeared only once in the dataset. While this improves consistency, it may hide meaningful repeated measurements.

Lastly, missing values were excluded during calculations by applying `na.rm = TRUE`. This prevents computational errors.

To sum the impact of these changes up, the steps improved overall data quality and usefulness, however they did reduce some detail and have potential limitations.

3.2 Generate necessary variables

To improve the analysis, we made new variables from the original datasets.

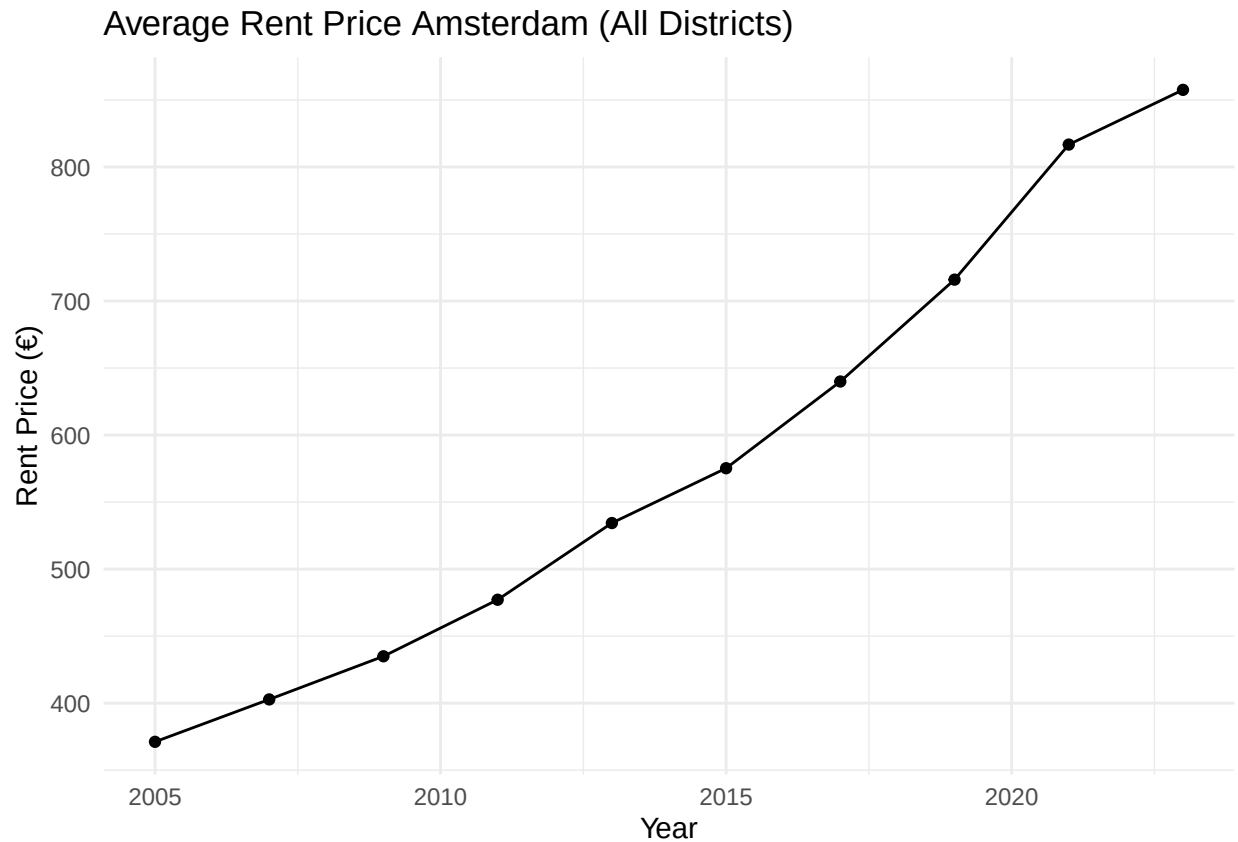
The variable **perc_studenten** was created by dividing the number of residents aged 15–25 by the total number of inhabitants in each neighbourhood and multiplying this by 100. This variable measures the percentage of young adults in a neighbourhood.

Second, we created **rondkomen_cat**. This variable groups the percentage of people who struggle to make ends meet into six categories. The categories range from **6.7% to 36.6%**. This makes it easier to show the data on a map.

Third, we created **studenten_cat**. This variable groups the percentage of young people into six categories, from **0% to above 20%**. This helps compare neighbourhoods more easily.

These new variables make the data easier to understand and easier to visualise. However, grouping values into categories means that some detailed information is lost.

3.3 Visualize temporal variation



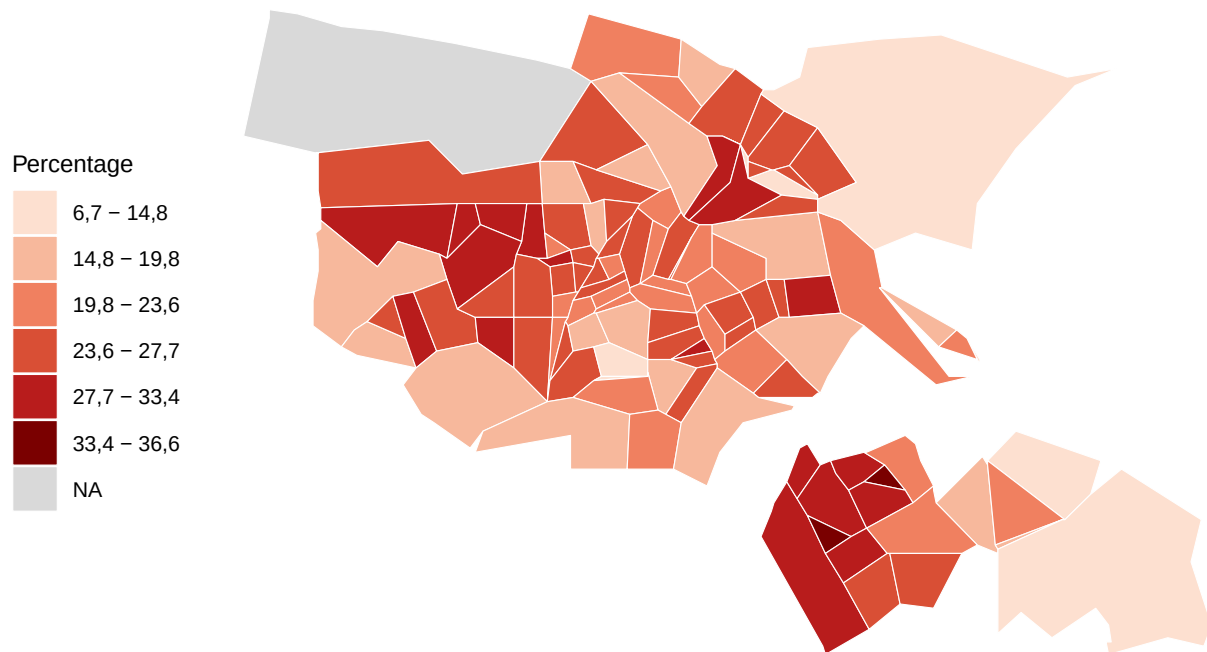
This graph shows how average rent in Amsterdam changed over the years. The trend is clearly upward. Rent more than doubled between 2005 and 2023. This matters because rising rent is one of the main drivers behind financial hardship. When rent grows faster than income people have less room left for other expenses. This graph shows why housing costs are such a big part of our social problem.

3.4 Visualize spatial variation

```
## Reading layer 'wijk_2023' from data source
##   'https://cartomap.github.io/nl/wgs84/wijk_2023.geojson' using driver 'GeoJSON'
## Simple feature collection with 3352 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: 3.358 ymin: 50.751 xmax: 7.227 ymax: 53.554
## Geodetic CRS:   WGS 84
```

Financial Hardship in Amsterdam, 2024

Per District, 18+



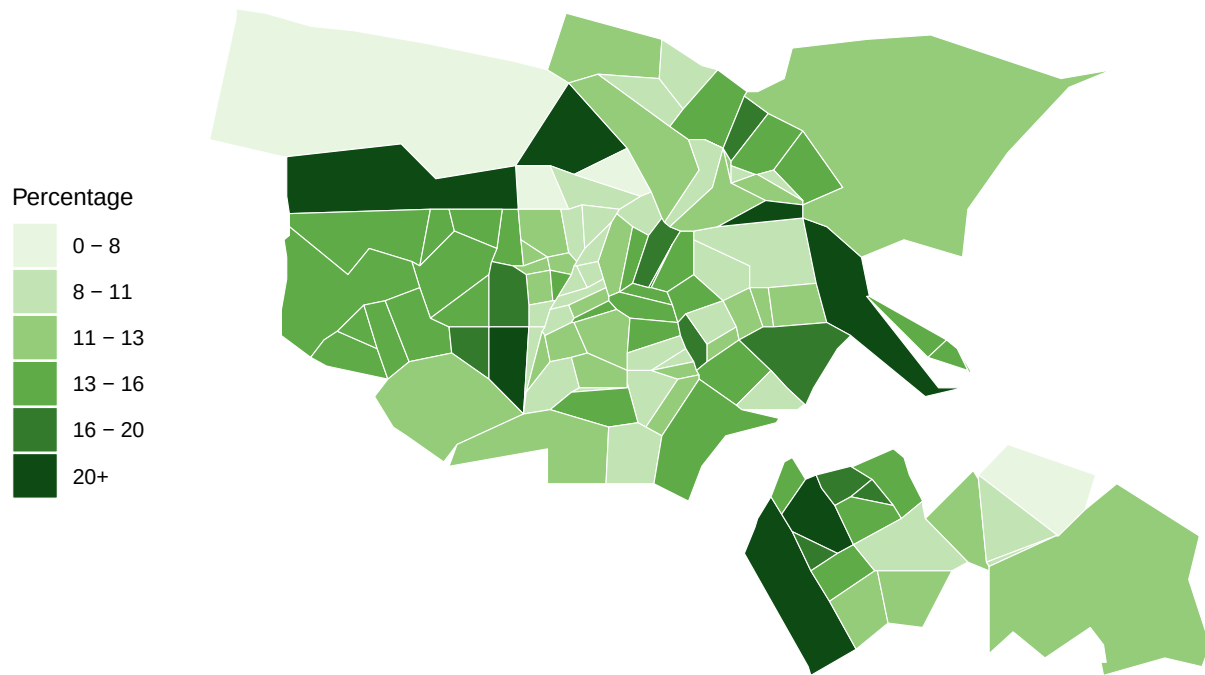
This map shows where financial hardship is highest across Amsterdam. Some districts clearly struggle a lot more than others. That already tells us something important. Money problems are not spread evenly across the city. It is probably connected to differences in housing costs and income between neighbourhoods. This map gives us a starting point to figure out where help is actually needed.

3.5 Visualize sub-population variation

```
## Reading layer 'wijk_2023' from data source
## 'https://cartomap.github.io/nl/wgs84/wijk_2023.geojson' using driver 'GeoJSON'
## Simple feature collection with 3352 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: 3.358 ymin: 50.751 xmax: 7.227 ymax: 53.554
## Geodetic CRS: WGS 84
```

Young adults/students in Amsterdam, 2024

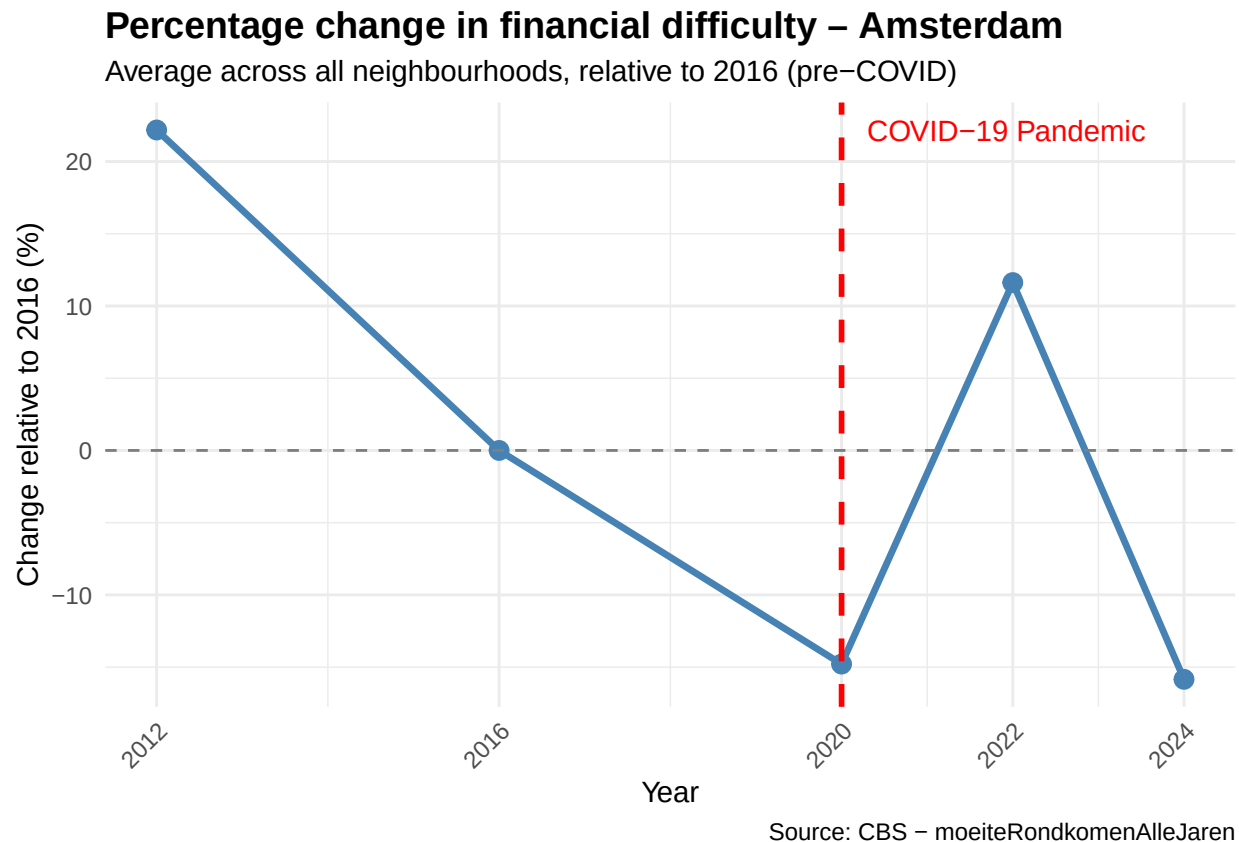
Percentage 15–25 years old per district



This map shows where young adults and students live across Amsterdam. We added this map because students are often seen as a financially vulnerable group. They usually have little or unstable income. They also pay relatively high rent compared to what they earn. This map lets us check which districts have the youngest population. That sets up the comparison with the hardship map in the next part.

When we compare both maps a pattern starts to appear. Districts with a high share of young adults often also show higher financial hardship. This suggests student neighbourhoods are not just young but also financially squeezed. It fits the bigger picture of our social problem. Housing costs in Amsterdam hit younger residents especially hard. So age and financial struggle seem to move together in several parts of the city.

3.6 Event analysis



This graph shows the yearly percentage change in financial hardship over time. The dashed line marks 2020 when COVID-19 hit. Around that period we see a noticeable shift in the trend. This is relevant because the pandemic affected jobs and income security for a lot of people. It helps show that financial hardship is not just driven by slow structural change. Sudden shocks like COVID-19 can also push more people into financial difficulty.

Part 4 - Discussion

4.1 Discuss your findings

The findings indicate that financial hardship in Amsterdam is closely related to rising living costs and differences in income across neighbourhoods. The rental price data show a strong increase in average rents between 2005 and 2023, with average rents more than doubling during this period. This suggests that housing has become significantly less affordable, especially for households whose incomes have not increased at the same pace.

The income dataset reveals substantial differences in average household income between Amsterdam neighbourhoods. Some neighbourhoods have considerably higher average incomes than others, indicating that residents do not experience rising living costs in the same way. For households in lower-income neighbourhoods, increasing housing costs are likely to have a greater impact on their financial situation because a larger share of their income must be spent on essential expenses.

The dataset on difficulties making ends meet provides further evidence that financial vulnerability remains an important issue in Amsterdam. When viewed together, the income, housing, and financial hardship data

suggest that residents with lower incomes are more likely to experience financial pressure. Rising rents, combined with increasing costs for food, energy, and other necessities, may leave these households with limited financial flexibility.

External factors should also be considered. The COVID-19 pandemic affected employment, income security, and economic activity, while the subsequent period of inflation increased the cost of living. These developments likely intensified financial difficulties for many residents and may help explain why a significant share of Amsterdam's population continues to struggle to make ends meet.

A limitation of this analysis is that it does not include all possible determinants of financial wellbeing, such as debt levels, savings, household composition, or educational attainment. Nevertheless, the combined evidence from the datasets suggests that rising housing costs and unequal income levels are key factors contributing to financial insecurity in Amsterdam.

Part 5 - Reproducibility

5.1 Github repository link

https://github.com/Wouter1334/programming4econ_groepje4

5.2 Reference list

Centraal Bureau voor de Statistiek. (2025, September 11). *Meer mensen tevreden over financiën in 2024, zorgen minder groot*. <https://www.cbs.nl/nl-nl/nieuws/2025/37/meer-mensen-tevreden-over-financien-in-2024-zorgen-minder-groot>

Mullainathan, S., & Shafir, E. (2013). *Scarcity: Why having too little means so much*. Times Books.

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